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3. A coupled partial melting Stokes-Darcy system. Strong form

$$\begin{aligned}
 (3.1) \quad & \tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
 & \mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0 \\
 & \nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi_f) \rho_r \mathbf{g} \\
 & \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0
 \end{aligned}$$

List SI units

q = pascal.

$\mathbf{v}$ = m/s

Non dimensionalization choice

- $\tilde{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0 \rho_r g_y / \mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$

The resulting non dimensionalized form is therefore

$$\begin{aligned}
 (3.2) \quad & \tilde{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \check{q}_f) = 0 \\
 & \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0 \\
 & \nabla \check{q} - \nabla \cdot \hat{\boldsymbol{\sigma}}(\check{\mathbf{v}}_s) = -(1-\phi_f) \rho_r \hat{\mathbf{g}} \\
 & \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0
 \end{aligned}$$

scaled nondimensionalized weak form let $\check{\boldsymbol{\sigma}} = \mathcal{D}\mathbf{v} - \frac{1}{3} \nabla \cdot \mathbf{v} \mathbf{I}$

$$\begin{aligned}
 (3.3) \quad & (2\phi_s \check{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
 & (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \check{q}_h, \omega \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_{f,h}, \omega \right) = 0 \\
 & (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r)) = 0 \\
 & (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s} \check{q}_{f,h}, \omega_f \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_h, \omega_f \right) = 0
 \end{aligned}$$

3.1. Variational form and discretization.

3.1.1. Stokes and variational formulation.

$$(3.4) \quad \begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(3.5) \quad \begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_{\Omega} - (p, \nabla \cdot \mathbf{v})_{\Omega} &= (\mathbf{f}, \mathbf{v})_{\Omega} + \langle \boldsymbol{\tau}^T \nabla \mathbf{u} \mathbf{v}, \mathbf{v} \cdot \boldsymbol{\tau} \rangle_{\Gamma_i} \\ &+ \langle \mathbf{v}^T \nabla \mathbf{u} \mathbf{v} - p_0, \mathbf{v} \cdot \mathbf{v} \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

let stress $\mathbf{s} \cdot \boldsymbol{\tau} = \boldsymbol{\tau}^T \nabla \mathbf{u} \mathbf{v}$ and $\mathbf{s} \cdot \mathbf{v} = \mathbf{v}^T \nabla \mathbf{u} \mathbf{v} - p_0$, we have traction boundary condition term defined on Γ_i which is $\langle \mathbf{s}, \mathbf{v} \rangle_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$(3.6) \quad \begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

3.2. Darcy.

$$(3.7) \quad \begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(3.8) \quad \begin{aligned} (\mathbf{u}, \mathbf{v})_{\Omega} - (p, \nabla \cdot \mathbf{v})_{\Omega} &= (\mathbf{g}, \mathbf{v})_{\Omega} - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

3.2.1. A degenerate Darcy test problem.

$$(3.9) \quad \begin{aligned} \mathbf{v} + \phi \nabla (\hat{\phi}^{-1/2} p) &= \mathbf{f} \\ \hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g \end{aligned}$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$(3.10) \quad \phi = \begin{cases} \cos^m(n(x^2 + y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases}$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$(3.11) \quad \phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases}$$

Define an arbitrary velocity and pressure

$$(3.12) \quad \begin{aligned} \mathbf{v} &= \left[\frac{x}{5}, -y \right]^T \\ p &= \hat{\phi}^{1/2} \end{aligned}$$

The source term is calculated as follows:

$$(3.13) \quad f = \mathbf{v}$$

and the divergence term is:

$$(3.14) \quad \begin{aligned} g &= \phi^{-1/2} \phi_+ \nabla \cdot \left[\frac{x^5}{5}, -x^4 y \right]^T \\ &= \phi^{-1/2} \phi_+ (x^4 - x^4) = 0 \end{aligned}$$

a divergence free test case

3.2.2. Modified BR element. The Bernardi-Raugel element has been researched extensively in previous literature [2]. The error estimate is well known established by [4].

$$(3.15) \quad \|\mathbf{u} - \mathbf{u}\|_{1,\Omega} + \|p - p_h\|_{0,\Omega} \leq Ch^m (\|\mathbf{u}\|_{m+1,\Omega} + \|p\|_{m,\Omega})$$

We state that our new function space satisfies the following hypotheses such that the error estimate holds:

- For any q in $H^m(\Omega) \cap L_0^2(\Omega)$, one has

$$\inf_{q_h \in \mathbf{M}_h} \|q - q_h\|_{0,\Omega} \leq Ch^m \|q\|_{m,\Omega}$$

- there exists a linear operator Π_h from $H^{m+1}(\Omega)^d \cap H_0^1(\Omega)^d$ into X_h such that

$$\forall \mathbf{v} \in H^{m+1}(\Omega)^d \cap H_0^1(\Omega)^d, \begin{cases} \forall q_h \in M_h, (q_h, \operatorname{div}(\mathbf{v} - \Pi_h \mathbf{v})) = 0, \\ \|\mathbf{v} - \Pi_h \mathbf{v}\| \leq Ch^m \|\mathbf{v}\|_{m+1,\Omega} \end{cases}$$

- for each q_h in M_h , there exists a function $\mathbf{v}_h$ in X_h such that

$$(\operatorname{div} \mathbf{v}_h, q_h) \geq \beta \|q_h\|_{0,\Omega} \|\mathbf{v}_h\|_{1,\Omega}$$

, where $\beta > 0$ is a constant independent of h .

We introduce a family of quadrilateral mesh of domain Ω , bounded by h .

4. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$(4.1) \quad \begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}$$

This is a double saddle point system. Symmetrically permute original linear system and form an enlarged saddle point system.

$$(4.2) \quad \begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}$$

(4.3)

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\mathbf{A}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [3], [6].

Consider the linear system of equation:

$$(4.4) \quad \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}$$

Algorithm 4.1 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$
-

In the previous literature [6], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works well with the stokes part. However, it converges very slow with the same approximation when applied to the Darcy part. We need a better preconditioner approximating inverse of the corresponding schur complement. We use MINRES algorithm [1] due to schur complement being symmetric and indefinite.

Algorithm 4.2 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Obtain $\mathbf{r} = \mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
 - 4: Obtain $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{y} \in V} \|(\mathbf{B}^T \mathbf{A}^{-1} \mathbf{B} - \mathbf{C})^{-1} \tilde{\mathbf{y}} - \mathbf{r}\|$
 - 5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
-

Based on the above algorithm, we proposed a new block preconditioner for our coupled linear system.

$$(4.5) \quad \mathbf{Q}^{-1} = \begin{bmatrix} \mathbf{S}_s^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d^{-1} \end{bmatrix}$$

where the schur complement matrices are approximated with MINRES method.

4.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

4.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [9]

5. Test cases. We test the linear system a little bit different from the linear system above.

5.1. Constant porosity and balanced pressure (divergence free velocity field).

5.2. Const porosity and unbalanced pressure.

$$(5.1) \quad \check{q} = 0 \quad \check{q}_f = 2(x+y) \quad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \quad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

$$(5.2) \quad \nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f} (x+y)$$

$$(5.3) \quad \mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \quad (\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I} = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix}$$

$$(5.4) \quad \begin{aligned} \check{\mathbf{v}}_r + \phi_f^{1/2} \nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2} \nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f} \check{q}_f &= \frac{-2}{1-\phi_f} (x+y) + \frac{1}{1-\phi_f} 2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} \check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f} (x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f} (x+y) = 0 \\ -2(1-\phi_f) \nabla \cdot \boldsymbol{\sigma}(\check{\mathbf{v}}_s) &= -2(1-\phi_f) \nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I}) = -\frac{8\phi_f^{1/2}}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

6. Retrieve of physical variables. Eventually, we need to retrieve actual physica values from nondimensionalized scaled variables. We need original velocity and pressure.

As stated above, the nondimensionalized scaled pressure potential $\check{q} = \frac{q}{\rho_r g_y l_0}$, hence $q = \rho_r g_y l_0 \check{q}$. The pressure potentials eliminated the effect of gravity, where $q_f = p_f - \rho_f g z$ and $q_s = p_s - \rho_f g z$. We can retrieve of original velocity with the similar manner $\check{\mathbf{v}} = \phi_f^{-1-\Theta} \mathbf{v}_r$.

7. Magma migration. For our very first simulation, we would like to simulate the production of magma in a simplified controlled scenerio. I was inspired by the setup described previously in[5]. The simulation will be setup in a chunk of rock ascending with constant upwelling velocity. A linear adiabatic temperature profile is applied here. A linear intial rock composition is also applied here.

7.1. Mathematical model. We are going to show our coupled darcy-stokes semi-degenerate model can solve this problem well. In[5], matrix creeping (Stokes equation) did not directly involved in the simulation due to the fact that boundary condition restricts a constant background upwelling ascending velocity. The solid matrix pressure is therefore strictly lithostatic pressure relating only to depth. In[5], only energy transports while concentration does not participate in transport phenomenon.

$$\begin{aligned}
 \check{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \check{q}_f) &= 0 \\
 \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) &= 0 \\
 \nabla \check{q} - \nabla \cdot \hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f) \hat{\mathbf{g}} \\
 \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) &= 0
 \end{aligned}
 \tag{7.1}$$

For simplification, we won't include diffusion into our transport model. We drop the diffusion term here for the fact that diffusion coefficient is relatively small compared to convective phenomenon. On the other hand, eutectic melting has a constant concentration such that gradient of concentration is of zero value. Hence the transport of component 2 (opx) can be expressed in terms of mixture theory as:

$$\begin{aligned}
 \bar{C}_t + \nabla \cdot (\bar{C} \mathbf{v}) &= 0 \\
 \bar{C}_t + \nabla \cdot (\mathbf{v}_e^* \bar{C}) &= 0
 \end{aligned}
 \tag{7.2}$$

where $\mathbf{v}_e^*$ denotes effective veclocity $\mathbf{v}_e^* = \kappa \mathbf{v}_f + (1-\kappa) \mathbf{v}_s$ and $\kappa = \frac{\phi_f C_f}{\bar{C}}$ where C_f and $\bar{C}$ using values from previous time step. Enthalpy is transported in a similar manner.

$$\bar{\mathcal{H}}_t + \nabla \cdot (\bar{\mathcal{H}} \mathbf{v} - \bar{K} \nabla T) = 0
 \tag{7.3}$$

7.1.1. Boundary conditions. In order to solve the proposed darcy-stokes coupled equation, we need appropriate boundary conditions defined around the computational domain. A detailed description is presented as following:

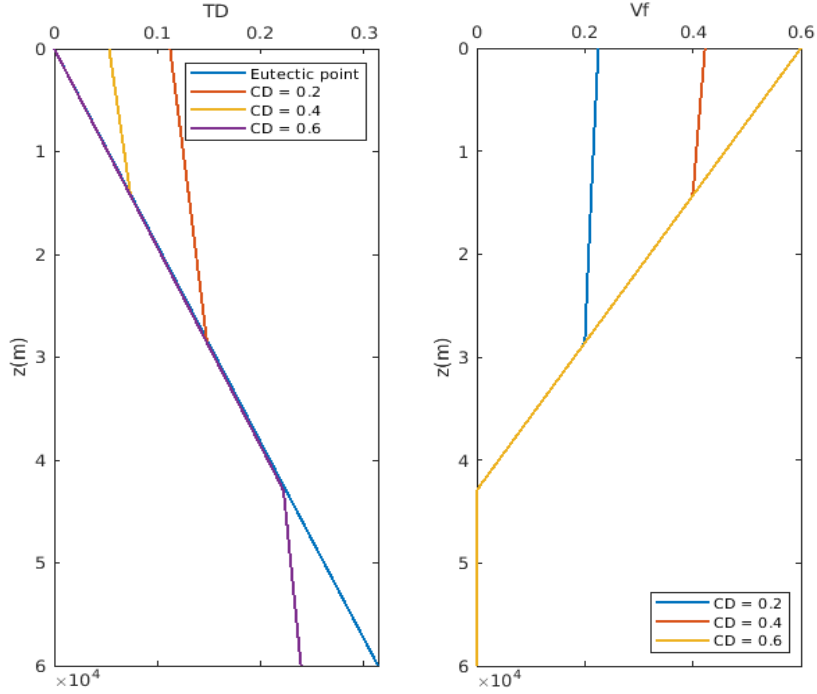
- Prescribe essential boundary condition of constant upwelling ascending velocity all around the computational domain. $\mathbf{v}_s = \mathbf{W}$ at no melting region.
- Prescribe natural boundary conditon at the place where melting happens.
- Prescribe essential (dirichlet) boundary condition of zero normal velocity on bottom, left and right sides.
- Prescribe natural boundary condition on top. A free outlet

7.1.2. Initial conditions. At the same time, we need an initial conditions for our transport equations. In[5], temperature and composition are distributed linearly along against the depth. Unlike [5], we implement eutectic melting model into our simulation. In our situation, temperature and total composition won't be deterministic. Here I make some modification on the setup. Initially, the entire domain was burried under the partial melting region, where

temperature was under the sub-solidus region. Melting starts when the entire chunk ascending adiabatically. As pressure continuously lowering, the eutectic point is lowering at the same time. Hence, melting flux supplies from top.

In eutectic melting situation, we cannot impose simple linear temperature directly due to the existence of the eutectic point. Therefore, an appropriate initial condition should be assigned with careful selection.

Linear Enthalpy distribution (Near constant).



7.1.3. Melting depth. As the magma chunk ascending adiabatically, we can define a depth z_m where melting starts. From the plots above, we can conclude that in our model

- Change of dimensionless composition can only affect the separation of eutectic and super-eutectic region.
- Change of dimensionless composition will not affect where magma leave eutectic melting region.

A fixed melting depth can therefore be calculated despite dimensionless composition distribution. The melting happens when current temperature reaches the eutectic melting temperature. In the situation of linear dimensionless enthalpy distribution $f(HD) = HD_0 + \eta z$. Assuming lithostatic pressure distribution, the eutectic point at given depth z is: $\mathcal{T}_e = 0 + \gamma^* \rho_s g z$. The melting depth is therefore calculated as: $z = \frac{HD_0}{\rho_s g - \eta}$. In this case, the smaller initial dimensionless enthalpy HD_0 the shorter melting depth.

7.2. Interpolation of stokes pressure. We are using piece-wise constant L2 approximation space for stokes and darcy pressure. In order to evaluate point-wise pressure. We are going to present an interpolation using generated piece-wise constant pressure.

7.3. Approximation of physical values on the cell edges. In the process of simulation, plenty of physical values are defined on the gauss points of cell edges. Take κ for example,

where $\kappa = \frac{\phi_f c_f}{\bar{c}}$, two phase parameters ϕ_f and c_f are computed with phase package using reconstructed values on both side of the cell edges. The value of $\bar{c}$ are directly obtained by WENO reconstruction. Therefore, two different values κ_{In} and κ_{Out} will be obtained during this computation process, which will be used in the calculation of the numerical flux.

7.4. Nondimensionalization and reterive of ordinary physical variables. We do the simulation with nondimensionlized variables. The unit less variables should be converted back to proper units.

Dimensionless time

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